

NephroTrend: A Deep Learning Framework for Worsening Renal Function Monitoring in Heart Failure Patients

Specific read codes and SNOMED codes for heart failure

Read Code	Term	SnomedCTDescriptionId
G580000	Acute congestive heart failure	18472010
G1yz100	Rheumatic left ventricular failure	72934016
G400.00	Acute cor pulmonale	82584011
G582.00	Acute heart failure	94251011
G41z.11	Chronic cor pulmonale	132655012
G58..00	Heart failure	139475013
G58..11	Cardiac failure	139482012
G581.00	Left ventricular failure	141306010
G580100	Chronic congestive heart failure	147247018
G580.12	Right heart failure	206703015
662T.00	Congestive heart failure monitoring	216184014
G5yy900	Left ventricular systolic dysfunction	216207010
G580200	Decompensated cardiac failure	300179017
G580300	Compensated cardiac failure	300180019
G581000	Acute left ventricular failure	300190010
Q48y100	Congenital cardiac failure	316833010
G554011	Congestive obstructive cardiomyopathy	350413012
G581.13	Impaired left ventricular function	411506018
Q490.00	Neonatal cardiac failure	412678013
8B29.00	Cardiac failure therapy	451426015
G580.13	Right ventricular failure	490972013
8HBE.00	Heart failure follow-up	1484917012
662W.00	Heart failure annual review	1484918019

1O1..00	Heart failure confirmed	1488804017
G5yyA00	Left ventricular diastolic dysfunction	1489358014
9On0.00	Left ventricular dysfunction monitoring first letter	2533628012
9On1.00	Left ventricular dysfunction monitoring second letter	2533629016
9On2.00	Left ventricular dysfunction monitoring third letter	2533630014
9N2p.00	Seen by community heart failure nurse	2548316014
9On3.00	Left ventricular dysfunction monitoring verbal invite	2549089012
662h.00	New York Heart Association Classification - Class III	2616472016
662i.00	New York Heart Association Classification - Class IV	2616473014
679X.00	Heart failure education	2645623019
G580400	Congestive heart failure due to valvular disease	2675255018
8Hk0.00	Referred to heart failure education group	311561000000117
8CMK.00	Has heart failure management plan	1746171000000120
G580.00	Congestive heart failure	70653017
G58z.00	Heart failure NOS	139475013
G580.11	Congestive cardiac failure	70653017
G21z100	Hypertensive heart disease NOS with CCF	107545013
G580.14	Biventricular congestive heart failure	153058012
G581.11	Cardiac asthma	119456014
9h1..00	Exception reporting - LVD quality indicators	1569101000000110
G211100	Benign hypertensive heart disease with congestive cardiac failure	299653017
9hH0.00	Excepted from heart failure quality indicators - patient unsuitable	1570021000000120
G581.12	Acute pulmonary oedema	492666016
8H2S.00	Emergency hospital admission for heart failure	3082850014
14A6.00	H/O: heart failure	251680018
SP11111	Heart failure as a complication of care	350484012
14AM.00	H/O: Heart failure in last year	453099015
1J60.00	Suspected heart failure	1488591011
9Or0.00	Heart failure review completed	308261000000111
662p.00	Heart failure 6 month review	404741000000119

	Heart failure monitoring - unstable symptoms	1539471000006110
	Heart failure monitoring - specialist clinical needs	1539481000006110
	Heart failure monitoring - social issues	1539491000006110
	Heart failure monitoring - psychological issues	1539501000006120
	Heart failure monitoring - multiple readmissions	1539511000006120
	Heart failure monitoring - co-medications	1539521000006110
	Heart failure monitoring - co-morbidities	1539531000006110
	Heart failure monitoring - palliative care	1539541000006120
	Cause of Death- Congestive Cardiac Failure	1576321000006110
G583.00	Heart failure with normal ejection fraction	2883808011
G583.11	HFNEF - heart failure with normal ejection fraction	1713091000000110
	Heart failure resolved	1728531000006120
	Heart failure monitoring in primary care	1856361000006120
	Heart failure monitoring in secondary care	1856371000006110
	AURAS-AF - consider the patient to have heart failure	1861731000006110
661N500	Heart failure self-management plan review	2115781000000110
661M500	Heart failure self-management plan agreed	2117931000000120
G583.12	Heart failure with preserved ejection fraction	2227501000000110
2JZ..00	On optimal heart failure therapy	2352391000000120
G58z.12	Cardiac failure NOS	139482012
33BA.00	Impaired left ventricular function	411506018
G584.00	Right ventricular failure	490972013
	CHF - Congestive heart failure	493289014
	Left heart failure	141303019
	Congestive cardiac failure	493287011
	LVF - Left ventricular failure	1235017018
	Impaired left ventricular func	1495323018
	Biventricular failure	510016018
	CCF - Congestive cardiac failure	493288018
	Heart failure with preserved ejection fraction	3496968011
	HF - Heart failure	1234906013
	Rehabilitation for heart failure	2170921000000110
	Heart failure self management plan	3515953019
	Symptomatic congestive heart failure	2974742013

	Systolic heart failure	2577903013
	Right heart failure secondary to left heart failure	73920011
	Cor pulmonale	138131012
	Congestive heart failure with right heart failure	2984250012
	Acute right-sided heart failure	474047012
	Heart failure due to end stage congenital heart disease	2950981012
	Acute exacerbation of chronic congestive heart failure	2973838013
	Diastolic heart failure	2577902015
	Exacerbation of congestive heart failure	2921293019
G210100	Malignant hypertensive heart disease with congestive cardiac failure	1236017010
G232.00	Hypertensive heart and renal disease with (congestive) heart failure	299672017
G234.00	Hypertensive heart and renal disease with both (congestive) heart failure and renal failure	299674016
	Decompensated chronic heart failure	2645367010
	High output heart failure	17613015
	Heart failure monitoring	2251951000000110
	Heart failure medication review	2956236014
	Heart failure with reduced ejection fraction	3008211011
	Chronic left-sided heart failure	178503015
	Chronic systolic heart failure	2816275019
	Chronic right-sided congestive heart failure	111275015
	Congestive heart failure due to left ventricular systolic dysfunction	2675256017
	Heart failure with reduced ejection fraction due to heart valve disease	3008139015
	Chronic left-sided congestive heart failure	9994016
	Chronic heart failure	80720010
	Heart failure with reduced ejection fraction due to coronary artery disease	3008136010
	Heart failure with reduced ejection fraction due to cardiomyopathy	3008202011
	Chronic right-sided heart failure	17997015
	Chronic diastolic heart failure	2816764017
	Heart failure with reduced ejection fraction due to myocarditis	3008268013
G58z.11	Weak heart	139475013
G557100	Beriberi heart disease	498953012
G554000	Congestive cardiomyopathy	1778488011
585f.00	Echocardiogram shows left ventricular systolic dysfunction	2159197017

G5yyE00	Right ventricular systolic dysfunction	2548656015
662f.00	New York Heart Association Classification - Class I	2616470012
662g.00	New York Heart Association Classification - Class II	2616471011
G5yyD00	Left ventricular cardiac dysfunction	2694523019
8CMW800	Heart failure clinical pathway	2205951000000120
G5y4z00	Post cardiac operation heart failure NOS	300214014
R2y1000	Cardiorespiratory failure	2472092014
8CeC.00	Preferred place of care for next exacerbation of heart failure	1765351000000110
388D.00	New York Heart Association Classification class	3637504017
8HTL.00	Referral to heart failure clinic	216246012
9N0k.00	Seen in heart failure clinic	2549243014
8HHb.00	Referral to heart failure nurse	2549697018
9N6T.00	Referred by heart failure nurse specialist	303361000000111
8Hg8.00	Discharge from practice nurse heart failure clinic	303441000000116
9N4s.00	Did not attend practice nurse heart failure clinic	303861000000118
8HHz.00	Referral to heart failure exercise programme	3011242019
9N4w.00	Did not attend heart failure clinic	407181000000116
9m5..00	High risk of heart failure screening invitation	735971000000112
8IE0.00	Referral to heart failure education group declined	1693911000000110
8HgD.00	Discharge from heart failure nurse service	1705341000000110
8IE1.00	Referral to heart failure exercise programme declined	3011446018
8IB8.00	Referral to heart failure exercise programme not indicated	3011612011
	Emergency heart failure admission since last appointment	1747691000006120
	AURAS-AF - consider the patient not to have heart failure	1861741000006120
	AURAS-AF - more time needed to decide on heart failure status	1861751000006120
679W100	Education about deteriorating heart failure	2122191000000110
8HTL000	Referral to rapid access heart failure clinic	2256811000000110
	CHADS2 (congestive heart failure, hypertension, age 75 years or older, diabetes mellitus and previous stroke or transient ischemic attack) score	3637564019
	Excepted from heart failure quality indicators - service unavailable	2774561000000110
	QOF (Quality and Outcomes Framework) heart failure quality indicator-related care invitation	2779841000000110
	Family history of heart failure	2764081016
	CHA2DS2-VASc (congestive heart failure, hypertension, age 2, diabetes mellitus, stroke 2 - vascular disease, age, sex category) stroke risk score	1774641000000110

	Family history of congestive heart failure	2764062012
	Right heart failure due to pulmonary hypertension	2674125010
	CHA2DS2-VASc (congestive heart failure, hypertension, age 2, diabetes mellitus, stroke 2, vascular disease, age, sex category) score	3513886018
67D4.00	Provision of information about heart failure	2451451000000120
8CL3.00	Discussion about heart failure care plan	2253411000000120
9hH..00	Exception reporting - heart failure quality indicators	1568511000000110
G210000	Malignant hypertensive heart disease without congestive heart failure	60617018
9hH1.00	Excepted from heart failure quality indicators - informed dissent	1570031000000110
8H7o000	Fast track heart failure referral for transthoracic two dimensional echocardiogram	1681621000000120
38DE000	CHA2DS2-VASc (congestive heart failure, hypertension, age 2, diabetes mellitus, stroke 2, vascular disease, age, sex category) stroke risk score	3513891017
38DE.00	CHADS2 (congestive heart failure, hypertension, age 75 years or older, diabetes mellitus and previous stroke or transient ischaemic attack) score	3637565018
	QOF (Quality and Outcomes Framework) heart failure quality indicator-related care invitation using preferred method of communication	1628291000000120
	Acute on chronic right-sided congestive heart failure	3849969010
	Assessment using CHADS2 (congestive heart failure, hypertension, age 75 years or older, diabetes mellitus and previous stroke or transient ischaemic attack) score	3637537015
L09y200	Cardiac failure following abortive pregnancy	2966901017
68B6.00	Heart failure screen	1484901018
1I10.00	Heart failure excluded	1488629013
9Or..00	Heart failure monitoring administration	1565031000000110
9Or3.00	Heart failure monitoring first letter	1569521000000110
9Or4.00	Heart failure monitoring second letter	1569731000000120
9Or5.00	Heart failure monitoring third letter	1569171000000120
	Heart failure information starter pack provided	1539391000006120
	Heart failure pathway protocol not followed	1822161000006120
	Heart failure monitoring default	1856381000006110
8I98.00	Heart failure rehabilitation programme not available	2405871000000120
	Heart failure monitoring invitation	1555941000000120
9Or1.00	Heart failure monitoring telephone invitation	1570071000000120
9Or2.00	Heart failure monitoring verbal invitation	1570041000000110
	Heart failure initial assessment	2205021000000110
	Heart failure with mid range ejection fraction	3786234019

